



Analog IC Design Implementation and Verification
Weeks 14-16 – Physical Verification, Extraction and Chip Assembly

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Physical Verification DRC/LVS

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What Is Physical Verification?

Physical verification: A process used to confirm the design rules and connectivity of an IC layout design:

- DRC
- ERC
- LVS
- Antenna check
- Density check



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Physical Verification: DRC

DRC is the process of checking physical layout data against fabrication-specific rules specified by the foundry to ensure successful fabrication.

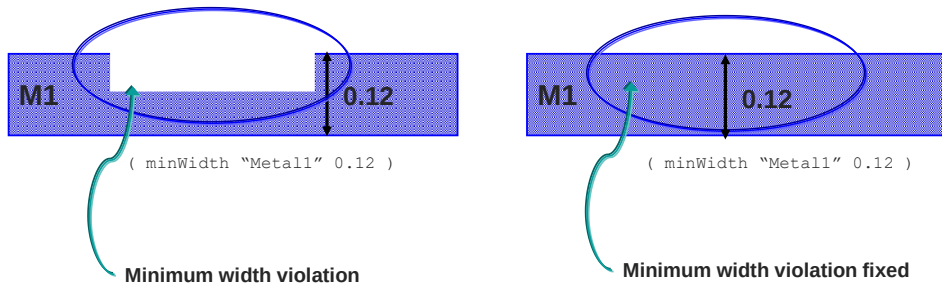
The following slides describe a few typical DRC errors and different ways of fixing them.



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DRC Example 1: Minimum Width

Minimum width (*minWidth*) : Specifies the minimum width of individual shapes on a single layer. Width is the difference between inside facing edges of individual shapes.

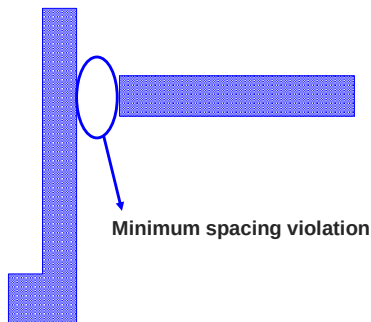


Left part of the slide has the violation.

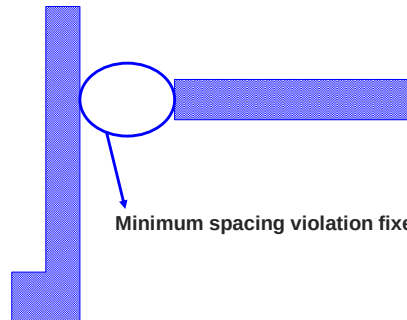
Right part of the slide has the violation fixed by increasing the spacing to meet the rule specified in the DRC Deck.

DRC Example 2: Minimum Spacing

Minimum spacing (*minSpacing*): Defines the minimum spacing between objects on a single layer. It can also be viewed as the shortest distance between two edges of different shapes of the same layer.



`(minSpacing "Metal1" 0.12)`



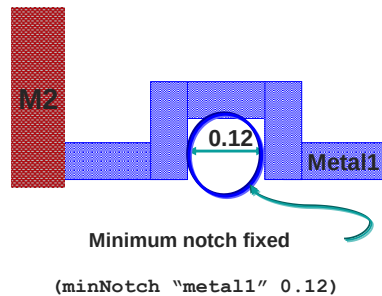
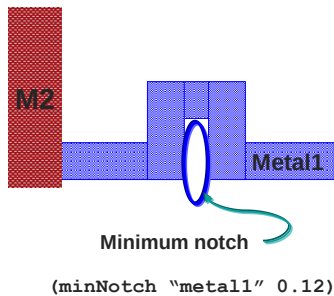
`(minSpacing "Metal1" 0.12)`

Minspacing: Minimum spacing between two similar layers:

1. metal to metal
2. via to via
3. diffusion to diffusion

DRC Example 3: Minimum Notch

Minimum same net spacing (*minNotch*): Specifies the minimum spacing rule for objects on the same net, including defining the minimum notch on a single-layer, merged object.



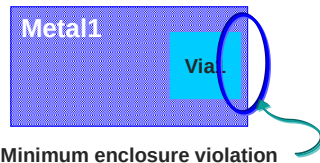
Min Notch: say for example: routers jog the wires to avoid congestion, etc.

A notch may be created.

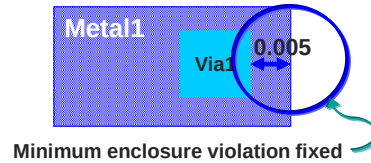
DRC Example 4: Minimum Enclosure

Minimum enclosure (*minEnclosure*): Specifies the enclosure of shapes on the second layer (via1) by shapes on the first layer (metal1).

- The enclosure is the distance between inside facing edges of shapes on the first layer to outside facing edges of shapes on the second layer.
- The *minEnclosure* rule implies that the second layer is fully enclosed by the first one.



`(minEnclosure "metal1" "Via1" 0.005)`



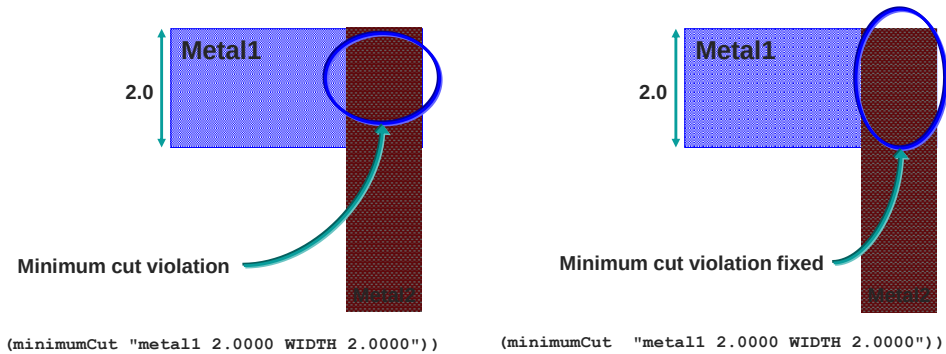
`(minEnclosure "metal1" "Via1" 0.005)`

Contact or via enclosure, has to be enclosed atleast by this particular area.

DRC Example 5: Minimum Cut

Minimum cut (*minimumCut*): Defines the minimum number of cuts a via must have when it is on a wide wire or pin whose width is greater than the width specified in the rule.

This rule applies to all vias containing the cut layer specified in the rule.



Design rule is given at the bottom from GPDK.

```
minimumCut "metal1 2.0000 WIDTH 2.0000") )
```

ERC

ERC checks are run to locate the following errors in layouts:

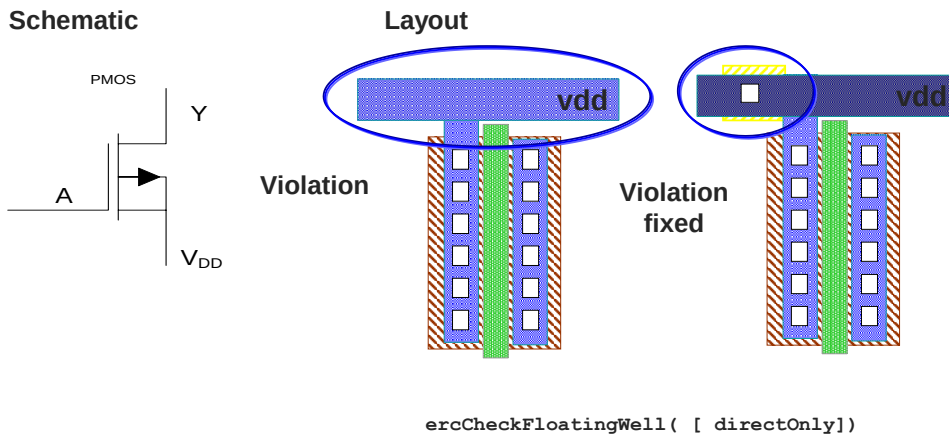
- Devices connected directly between power and ground
- Floating devices and floating wells
- Devices that are shorted
- Devices with missing connections



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ERC Error: Example 1

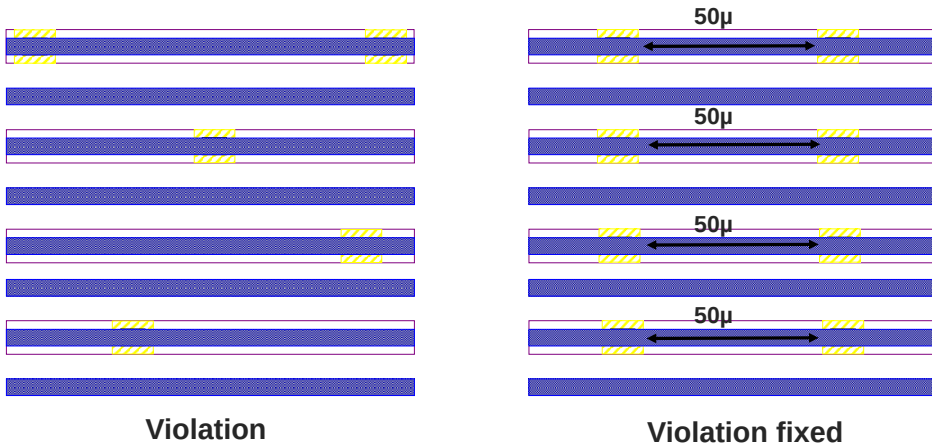
Well tap connection error: The well taps should bias the wells as specified in the schematics.



Ntap added in the solution, which was missing.

ERC Error: Example 2

Well tap density error: If there are not enough taps for a given area then this error is flagged. To prevent this kind of error, taps need to be placed regularly, which biases the well to prevent latchup. For example, in a typical 90 nm process, the well-tap-density rule requires well taps to be placed every 50 microns.



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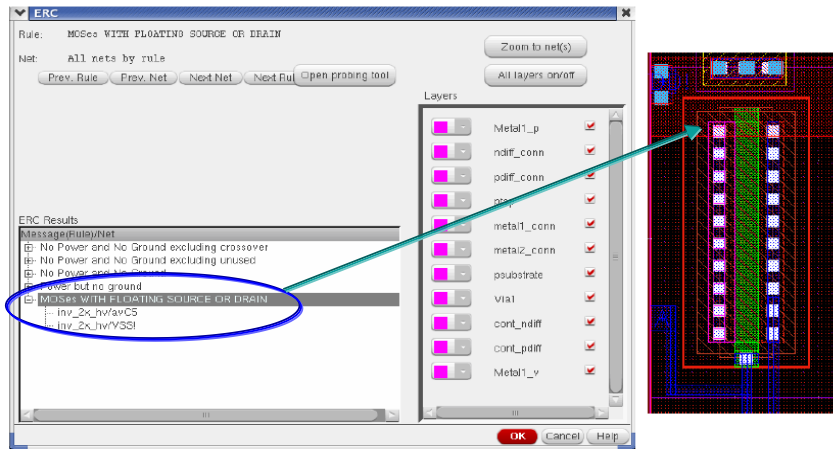


The above example shows the power rails.

1. With Violation >> Shows the irregular placement of well taps in an area.
2. With Violation Fixed >> Shows the placement of well taps in an area with respect to well tap density rule for a typical 90nm process.
3. This is an ERC rule. ERC rules are not specified in the gdpk library.

Unconnected Terminals Check

The ERC form shows that the source or drain of the PMOS device is floating.

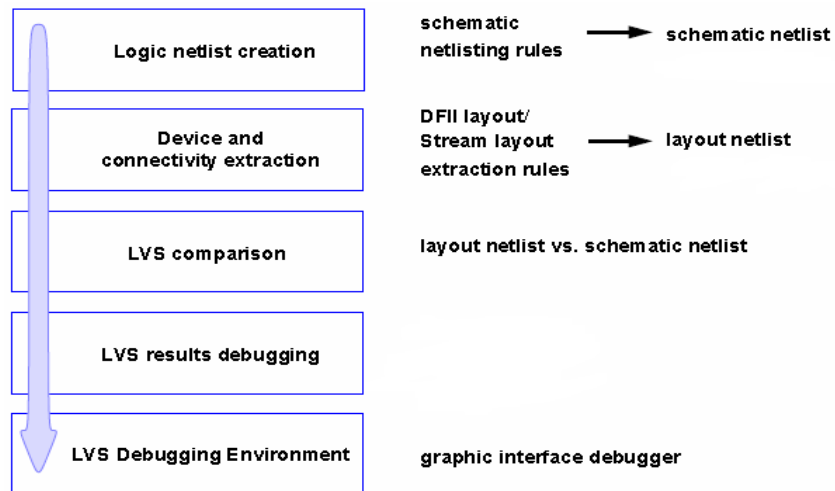


```
ercCheckFloatingDevices ( [resOneTerm] [resBothTerm] [capOneTerm] [capBothTerm]  
[mosSourceOrDrain] [mosSourceAndDrain] [mosGate])
```

The examples show a few of the ways the ERC checks can be configured to catch unconnected terminals of devices. The checks can also be configured to ignore devices that have not been connected at all.

Physical Verification: LVS

LVS: Comparing a physical layout design to the schematic from which it was designed.



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LVS Checks

Extract errors

- Shorts
- Opens

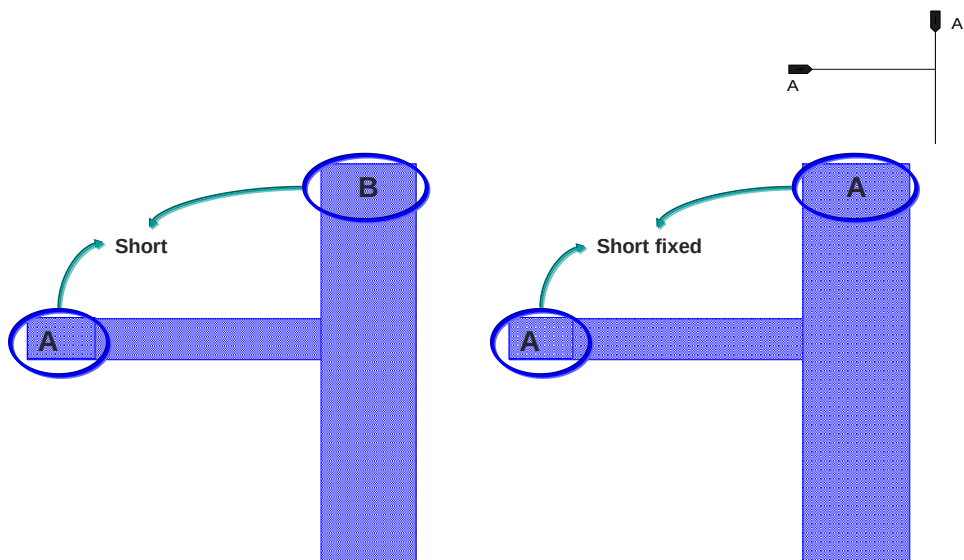
Compare errors

- Pin errors
- Parametric comparison
- Device mismatch
- Net mismatch
- Malformed devices



These are all flagged by the lvs tools If they are not matched with the schematic.

LVS: Extract Errors – Short



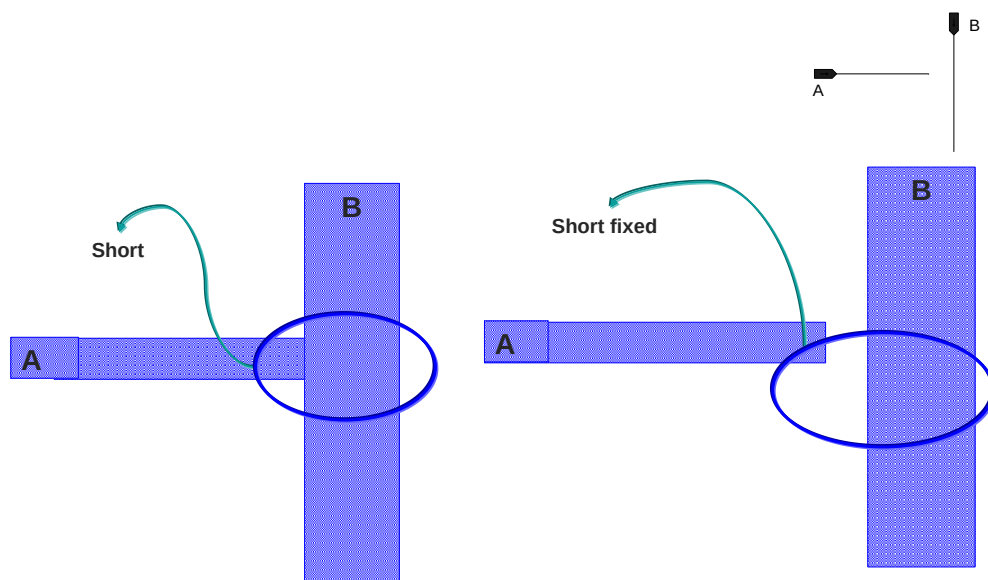
Same net named with different pin names

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A and B are shorted being on the same net “net1”.

LVS: Extract Errors – Short (continued)



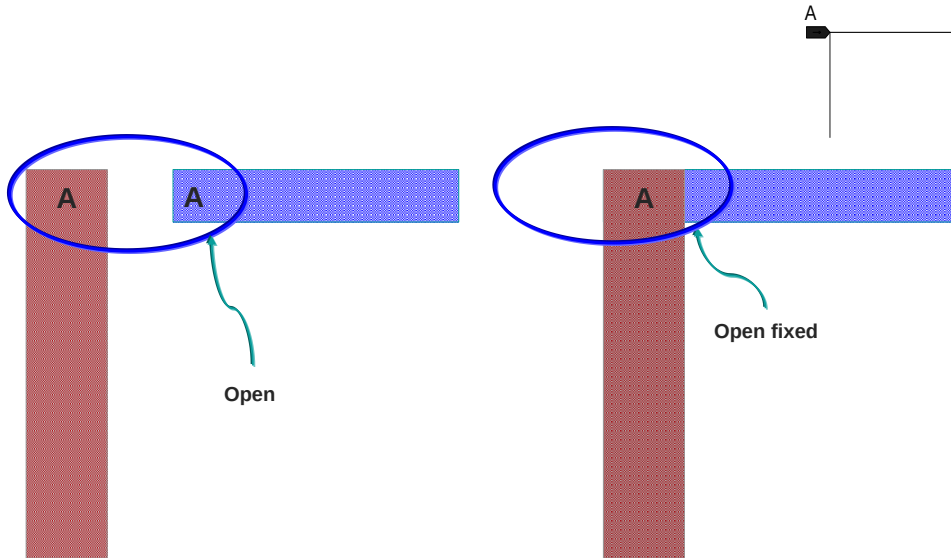
Two different nets shorting together

Two nets: n1 connected to Pin:A, n2 connected to Pin:B

Accidentally got shorted, during routing.

Left is a problem,, right is the fix.

LVS: Extract Errors – Open

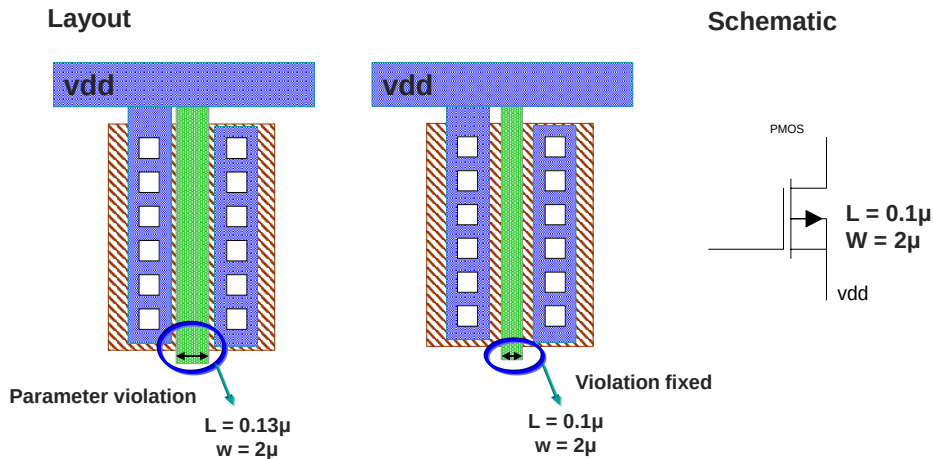


Same net routed in two different metal layers but not connected

This slide explains a text open error which occurs when there is no connectivity exists between a single net routed in two different metal layers.

LVS: Compare Errors – Parameter Mismatch

Device parameters on schematic and layout are compared. For example, with this transistor, LVS checks are necessary parameters such as width, length, multiplication factor, and so forth.

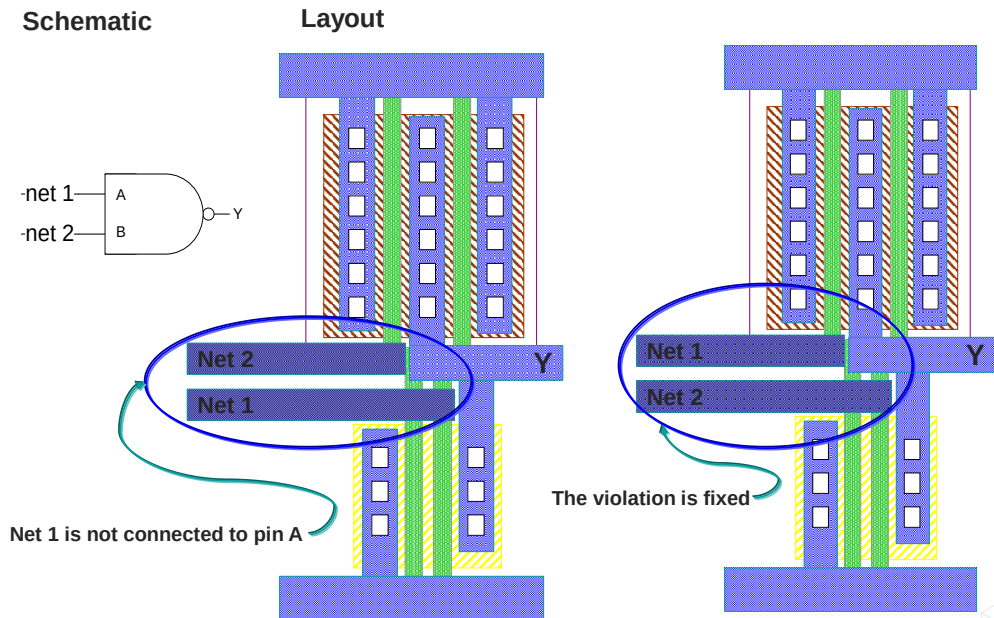


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Here is an example of a typical parameter mismatch shown. The width of the device is different in schematic and layout which will lead to parameter mismatch errors.

LVS: Compare Errors – Device or Net Mismatch

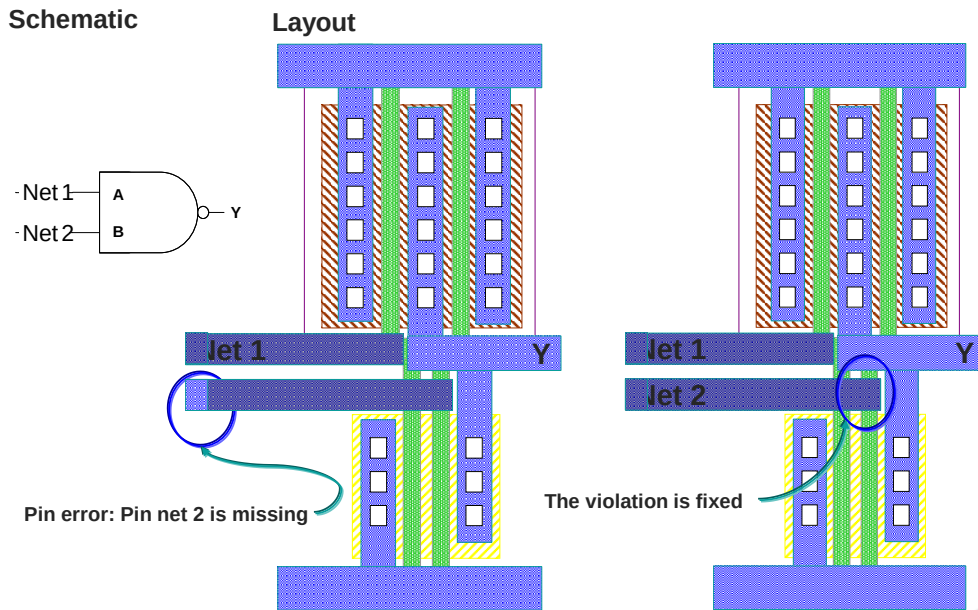


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Here we are discussing a device mismatch and net mismatch error. In the schematic net 1 is connected to pin A of nand2 and net 2 is connected to pin B of nand2, in the incorrect layout net 2 is connected to pin A and net 1 is connected to pin B. which is fixed in on the right-hand side layout. This leads to device mismatch and net mismatch errors.

LVS: Compare Errors – Pin Errors



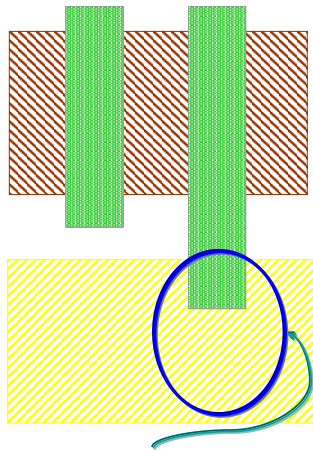
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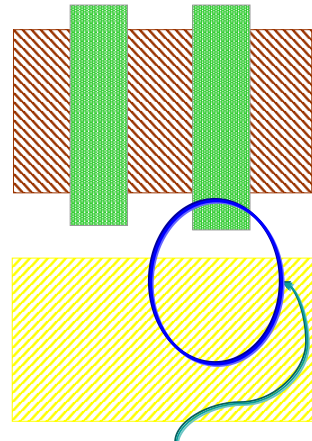
This slide explains an LVS pin error which occurs when pins on schematic and layout are not matching. In the figure nand2 schematic has two pins namely a and b but layout of nand2 has only pin a and pin b is missing which leads to a pin error.

LVS: Compare Errors – Malformed Devices

Example: Accidentally routed poly forms an abnormal device.



Malformed device violation



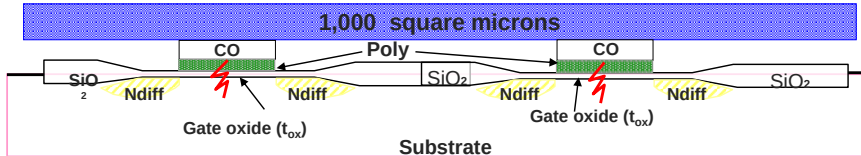
Malformed device violation fixed

Here we are explaining malformed devices which are created accidentally when a poly crosses diffusion. These devices are unintentional and should be removed from the design.

Antenna Ratio: Example

Antenna ratio = area of metal/area of gate

- Each foundry sets a maximum allowable antenna ratio for the chips they fabricate.
- For example, assume a foundry setting is a maximum allowable antenna ratio of 500. If a net has two input gates and each has an area of 1 square micron, any metal layers that connect to the gates and have an area larger than 1,000 square microns have process antenna violations because they cause the antenna ratio to be higher than 500.
- Antenna ratio = $\frac{1000}{1+1} = \frac{1000}{2} = 500$.



- Thin oxide is potentially damaged by the discharge of carriers through oxide.

This ratio can be seen in LEF files supplied by foundry.

Most of the routers have the option to automatically fix antenna (e.g., NanoRoute®).

Antenna Checks

- During deep submicron wafer fabrication, gate damage can occur when excessive static charges accumulate and discharge, passing current through a gate.
- If the area of the layer connected directly to the gate or connected to the gate through lower layers is large compared to the area of the gate and the static charges are discharged through the gate, the discharge can damage the oxide that insulates the gate and cause the chip to fail. Because this phenomenon occurs during the process, it is known as (PAE).
- To determine the extent of PAE, the router calculates the area of the layer compared to the area of the gates directly connected to it or connected to it through lower layers.



Here we are explaining process antenna effect and how it happens.

How to Identify and Fix Antenna Errors

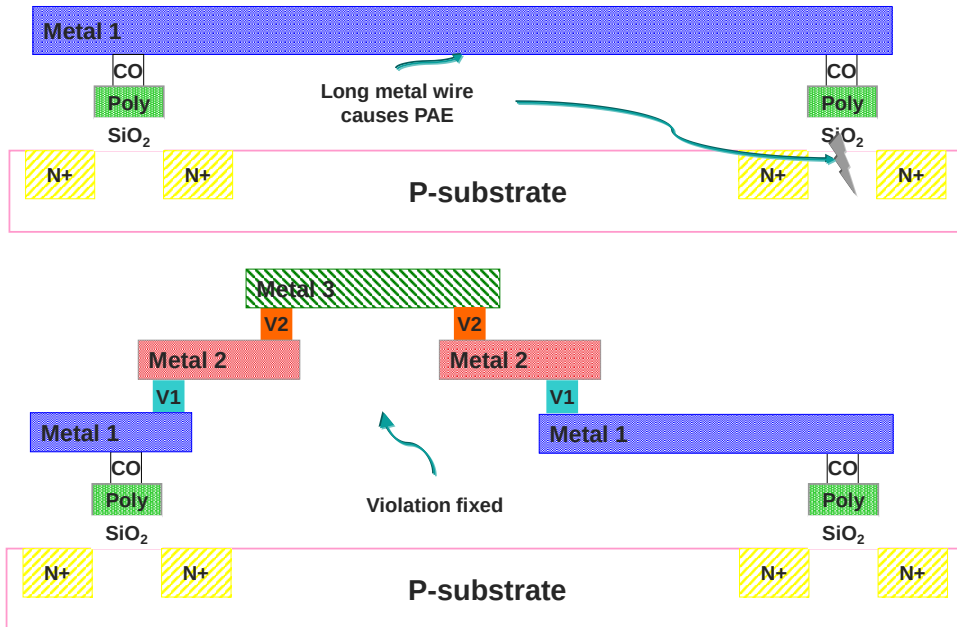
- Antenna checks are also called [node area checks](#).
- Antenna errors of a node can be fixed using the following techniques:
 - Connect an N-diode on the same net, which acts as a short resistance for electrostatic discharge.
 - Adjust the geometries of the metal layers of a net so that various area ratios do not exceed the area ratio limits.
- Some errors can be corrected using any one of the techniques and some might have only one valid fix:
 - Metal splitting or layer stitching
 - Diode insertion



Metal splitting is an easy method to fix PAE compared to diode insertion.

Diode insertion also increases area.

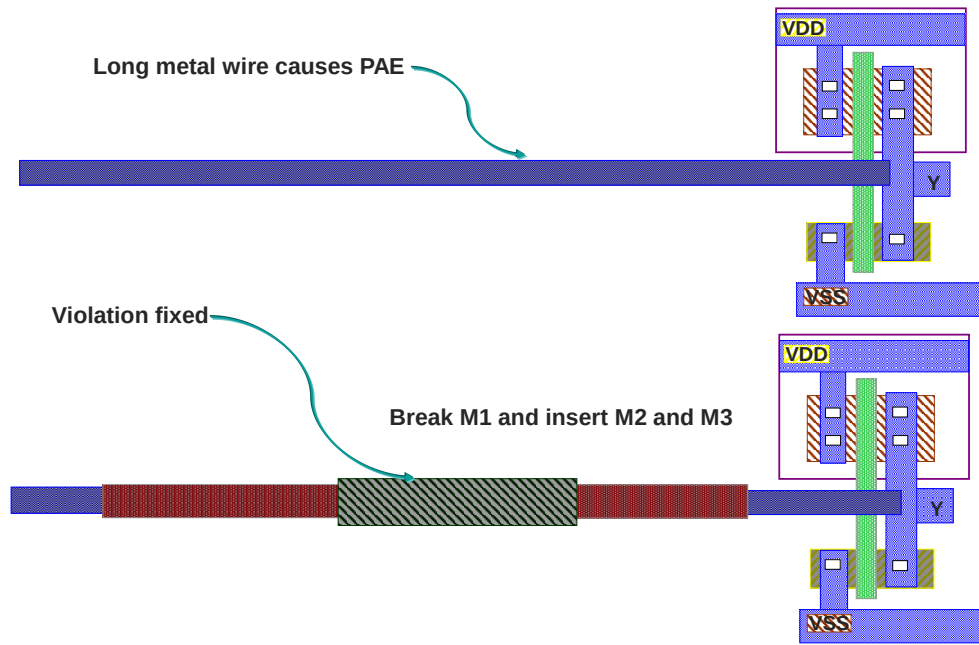
Metal Splitting



The top picture shows that a long metal wire causes the Process Antenna Effect (PAE). Because the area of the metal1 connected to gate damages the thin oxide under the poly.

The picture at the bottom shows the solution called metal splitting, where the long metal wire (metal1) is broken and connected using different metal layers.

Layer Stitching



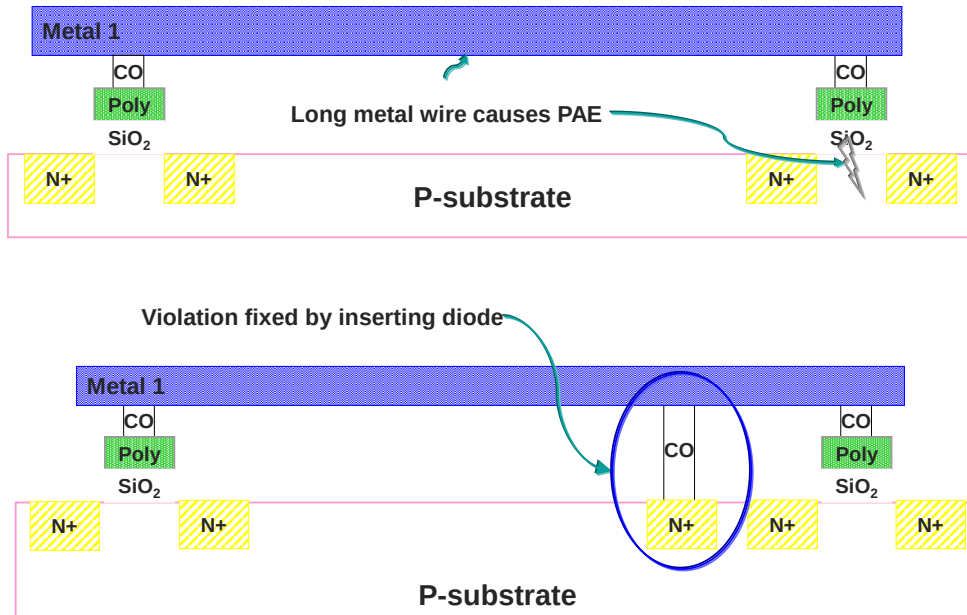
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Another approach for reducing metal 1 length by inserting metal 2 and metal 3 segments in between the long metal 1 net.

Diode Insertion

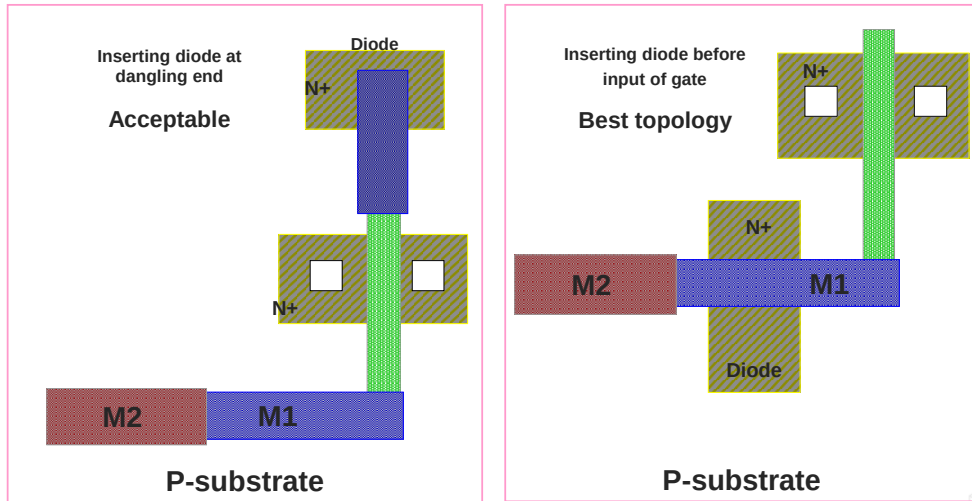


In the picture (bottom): The diode is formed between the P-substrate and the N+ and is connected to metal1.

The top picture has a long metal1 connecting to the gates, thus causing the violation.

Where to Insert the Diode

The best solution is to connect the diode at the connected end of the gate instead of the unconnected end of the gate, which has the effect of increasing the poly resistance.



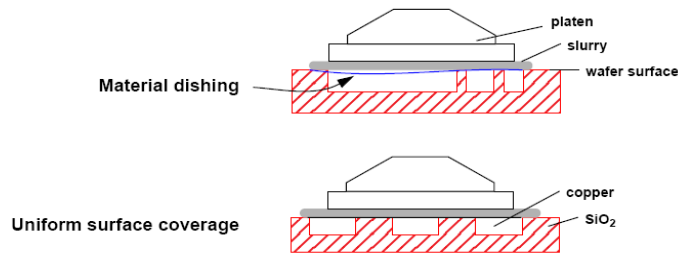
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In the previous slide we talked about typical fixes for antenna violation one of them was diode insertion. Here we explain the preferred location for placement of diode. A diode can be connected to unconnected end of gate but due to high poly resistance it is always advisable to connect diode to connected end of poly.

Density Checks

- Find areas of nonuniform layer coverage that erode during chemical mechanical polishing (CMP).
- Detect areas of potential material dishing, resulting in nonplanar surfaces.



Density checking is a process-related check concerning the uniformity of the wafer surface during Chemical Mechanical Polishing (CMP). During the CMP portion of damascene processing, lack of surface uniformity between copper and silicon dioxide may result in dishing or erosion, respectively. This dishing and erosion may increase interconnect resistance and coupling capacitance, resulting in degraded circuit performance.

The platen used during CMP provides a mechanical force that abrades a chemically reactive slurry on the wafer surface. This combination chemical/mechanical effect grows oxide on the surface of the wafer and simultaneously rubs off the softer, newly grown oxide. Oxide growth and subsequent abrasion rates are different for various materials. The uniformity of their respective coverage provides a substantially flatter wafer surface.

Density Rules

Minimum density rule (*minimumDensity*): Defines the minimum percent of the total design area that the metal in the specified layer must occupy.

$$\text{density} = \text{metal layer area} / \text{total design area}$$

The density for the specified metal layer must be greater than or equal to the minimum density specified in this rule, without violating the maximum density rule.

Maximum density rule (*maximumDensity*): Defines the maximum percent of the total design area the metal in the specified layer must occupy.

$$\text{density} = \text{metal layer area} / \text{total design area}$$

The density for the specified metal layer must be less than or equal to the maximum density specified in this rule without violating the minimum density rule.

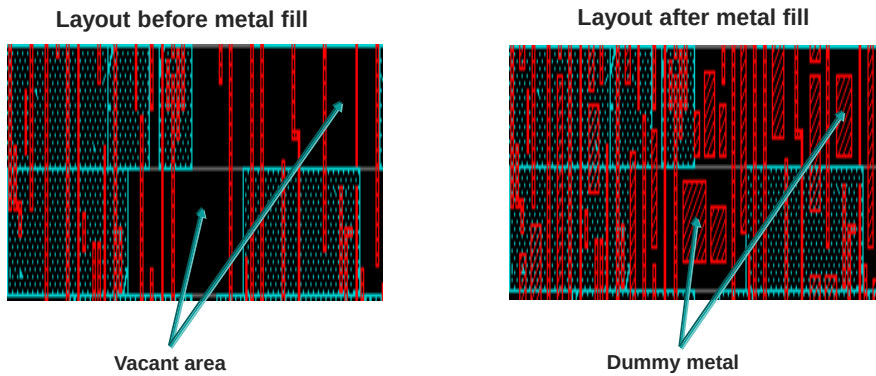


Min density rule should make sure “max density rule” is not violated.

Max density rule should make sure “min density rule” is not violated.

How to Fix Metal Density Violations

Add dummy metal in the vacant areas as shown in the example:



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Here we explain methods to fix density violations.

Density violations are not specified in the DRC rule decks.



Subsection

Formal Verification

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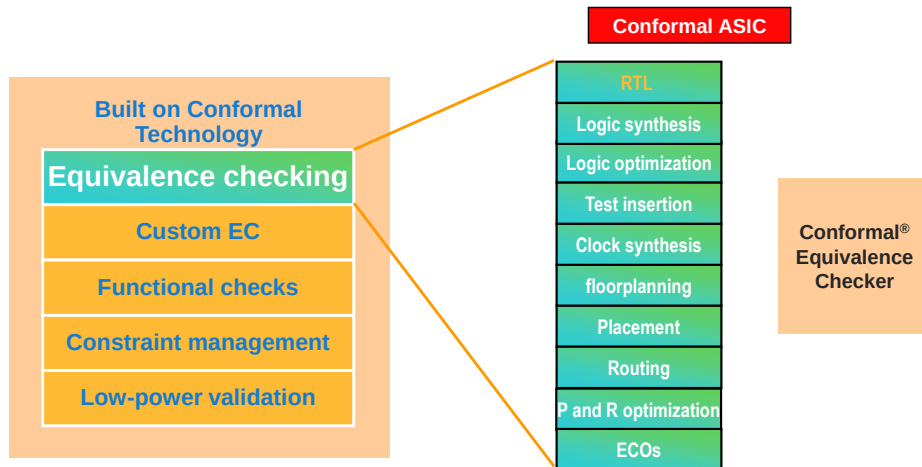
What Is Formal Verification?

Formal verification: A process of proving or disproving the correctness of the logic functionality between a golden (original) design and a revised (changed) design:

- Can also be described as *static functionality checking* of two designs (golden and revised) where no testbench or vectors are needed.
- Involves exhaustive exploration of all possible behaviors.
- Enables designers, by the orders of magnitude faster than dynamic verification, to quickly verify two copies of the designs in the typical place and route flow where the designs are transformed at different stages, post-place-optimization, clock-tree-synthesis, and so forth.

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Formal Verification: Digital Flow



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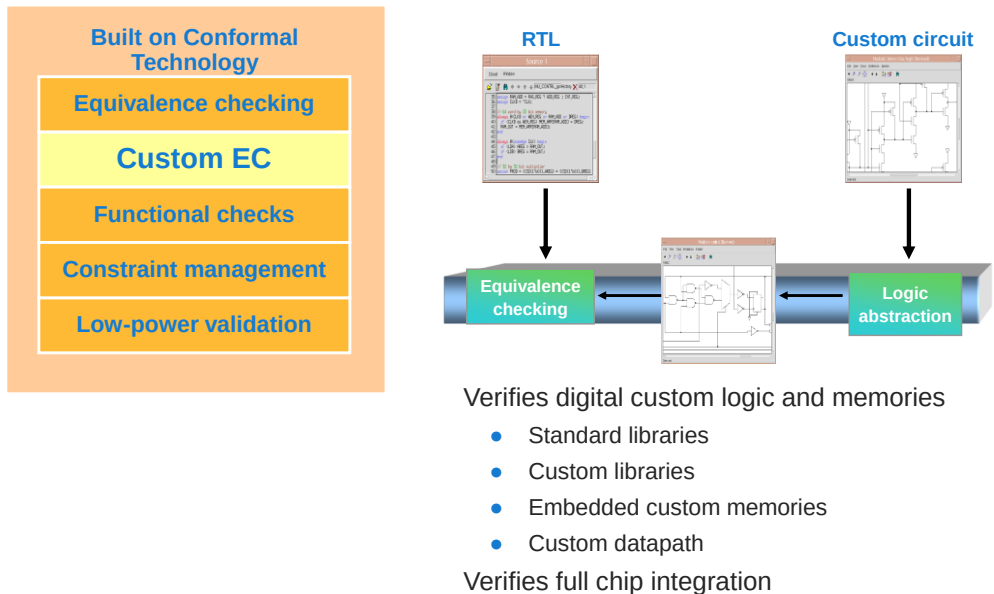


This slide shows the flow for a typical digital design scenario.

1. From RTL to GDS2 the design goes through several transformations and the verification needs to be done during every transformation to make sure the functionality remains the same.
2. For example: On the RHS flow, “Logic-optimization” is the golden and Test-Inserted netlist is the revised version. During test insertion scan

Flops are inserted and we need to make sure the scan-insertion process doesn’t change the functionality. Hence, there is a need to run Formal Verification between these two stages.

Formal Verification: Custom Flow



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1. This slide explains the formal verification flow in custom designs.
2. ADC or DAC can be an example of this, where the transistors can be converted one level abstraction up, to a gate level representation.
3. RTL is converted to one level abstraction down to a gate level representation.
4. Gates level representations of Golden and Revised are compared for functionality.
5. This can be used easily in a Mixed Signal IC flows where a part of the design is Custom.



Subsection

Extraction and Backannotation

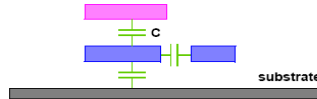
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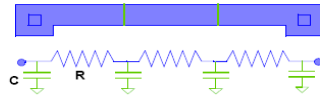
Parasitic Extraction

Extraction: The process of extracting parasitics from a design that includes device and interconnect parasitics.

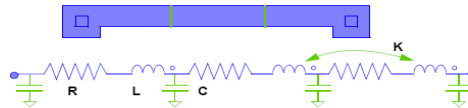
Basic parasitic capacitors



RC parasitic networks with distributed capacitors



RCLK parasitic networks



Here we discuss about parasitic resistance and capacitance extraction, different types of parasitic modeling.

Characteristics of a Good Extractor

It creates accurate electrical models of physical interconnect and devices:

- Deep submicron (DSM) hierarchical extraction.
- Full-chip interconnect capacity.
- Sophisticated data management for post-layout analysis.

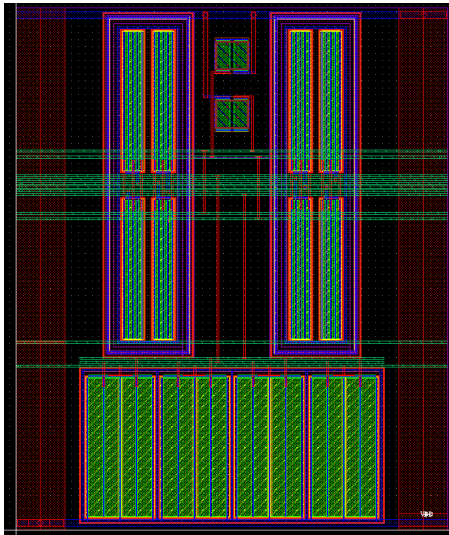
It accurately models all latest process manufacturing effects:

- Wire edge enlargements (WEE): Impact on resistance and capacitance can be modeled separately.
- Thickness variation (erosion and etching): Modeled as a function of metal width and effective density.
- Metal fill: Engine should handle both floating and grounded nets.
- Litho-aware extraction.

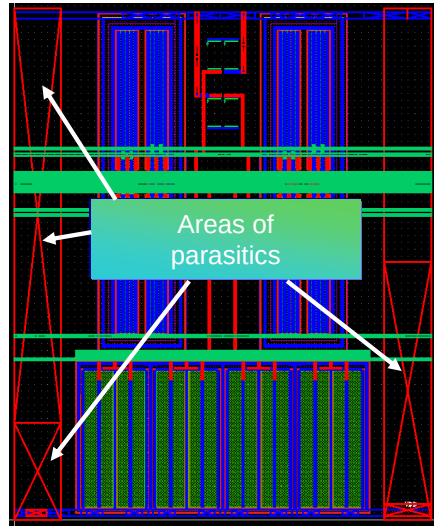


Typical features of an extractor.

Layout Extraction



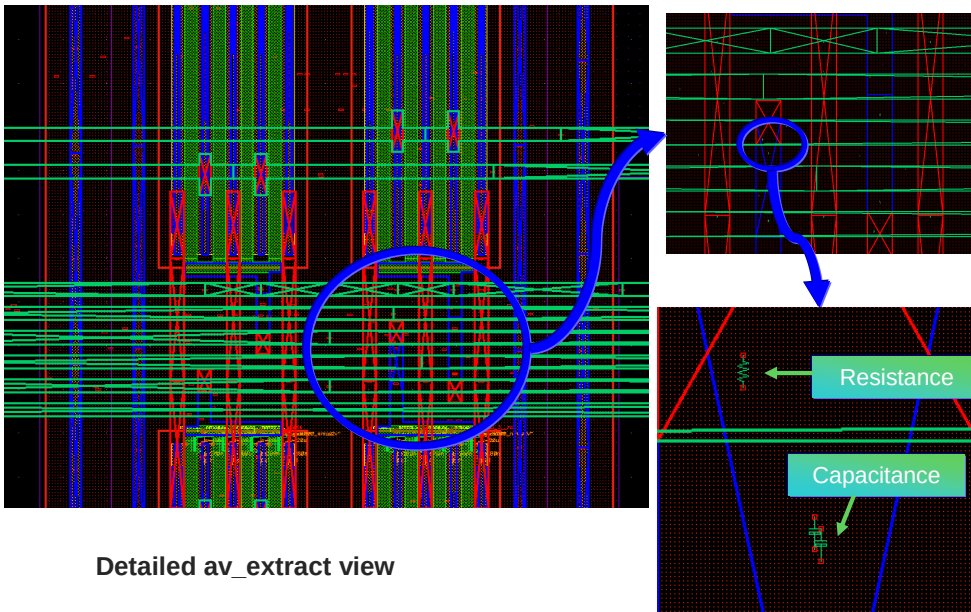
Layout view



Extracted view

Places to find the parasitics are the cross marked areas.

Layout Extraction (continued)



Detailed av_extract view

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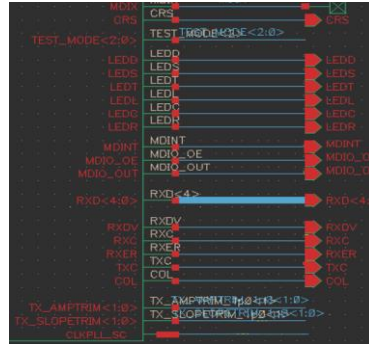
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Extracted view shows the parasitic symbols in the layout.

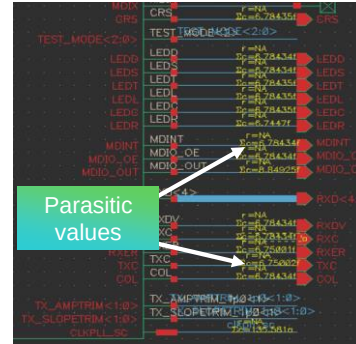
Parasitic Backannotation

Backannotation: The process of providing extracted parasitic data for analysis and optimization during the post-layout optimization phase.

The extracted parasitic information is stored in different file formats such as Distributed SPICE Parasitic Format (DSPF), SPICE (extracted SPICE netlist), and Standard Parasitic Extraction Format (SPEF).



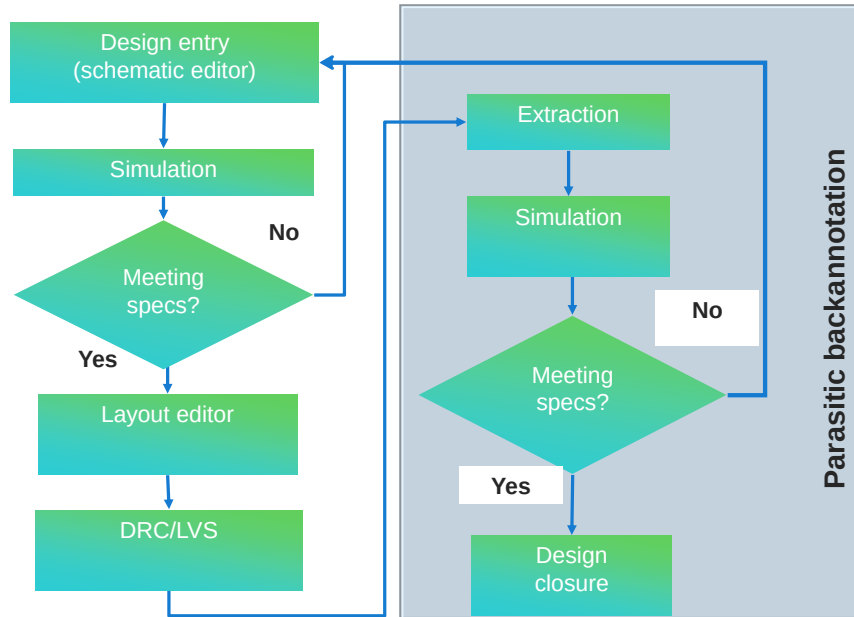
Before annotation



After annotation

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Analog Extraction and Backannotation Flow



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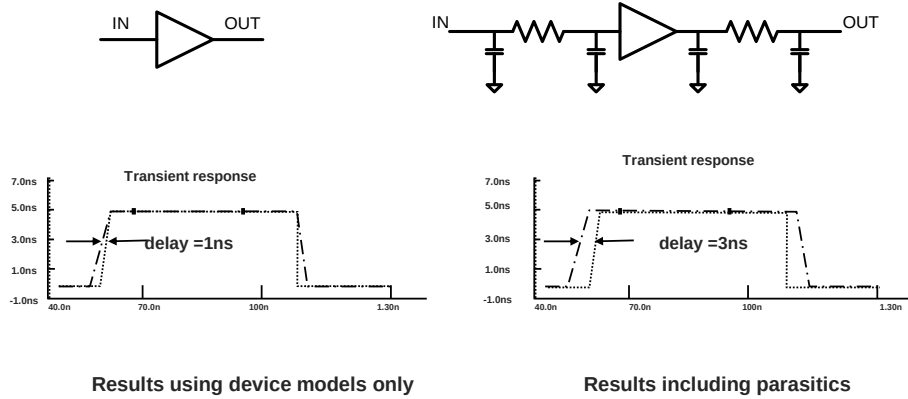
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This slide explains extraction and back annotation flow in analog environment.

1. The design entry is made in the schematic editor
2. The schematic is simulated and is verified with specifications.
3. The schematic which meets specifications is taken to layout editor for implementation.
4. The implemented layout is verified for DRC and LVS errors.
5. The verified layout is extracted and extracted parasitic data is back annotated in the form of extracted spice netlist for simulation.
6. If the design meets specifications, then the design is taken forward for design closure else optimizations are performed on the schematic until design meets specifications.

Post-Layout Extraction Simulation

Simulating parasitic elements as a part of the design is required to accurately model the manufactured circuit.



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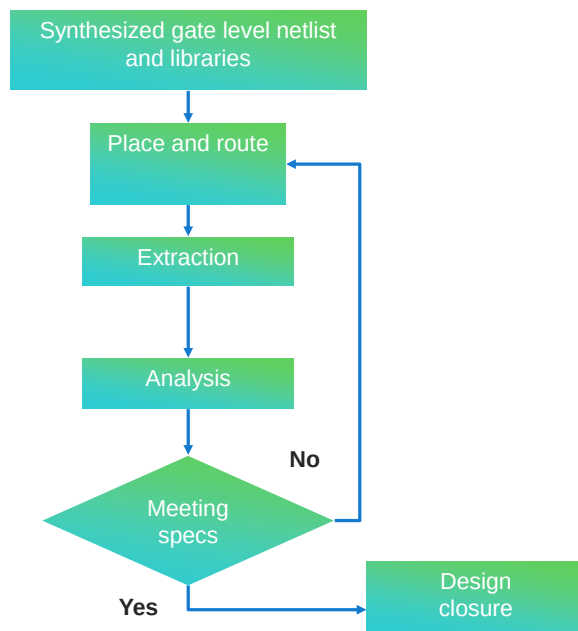
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The importance of parasitic extraction by showing the response of a circuit with and with out parasitic modeling.

Left one: Prelayout (schematic)

Right one: Post Layout (after annotation).

Digital Extraction and Backannotation Flow



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In digital extraction and backannotation only interconnects are extracted device extraction data is available in libraries.



Chip Assembly and Finishing

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Module Objectives

In this module, you learn how to

- Perform top-level assembly of a mixed-signal design (with both staggered and normal I/O pads) including swapping out abstracts with layout and GDSII, texturing I/O pads, routing with top-level devices using analog routers.
- Finish the chip by adding logos and version information.



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Chip Assembly Tasks

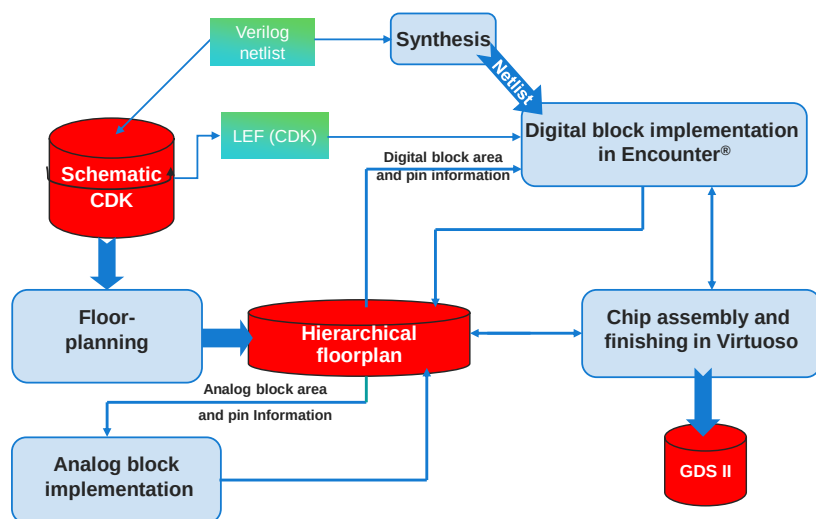
Chip assembly, as the name implies, is the process of combining different blocks of a chip with the I/O pad cells and integrating them all into a completed design.

- Chip-assembly tasks greatly depend on the full-chip-assembly methodology that is being used for the respective designs:
 - Analog on top (AOT)
 - Digital on top (DOT)
- Full chip assembly for AOT designs, which have a significant number of custom design blocks, is done using Virtuoso® Preview.
- In DOT designs, where there are fewer custom design blocks than digital place and routed blocks, the chip-finishing tasks are done by using SoC Encounter platform.



SOCE: Silicon On Chip Encounter

Analog-On-Top Design Flow



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Chip-Finishing Tasks: AOT

Chip-finishing tasks of AOT methodology comprise the following activities:

- Full-chip simulations
 - Full chip AMS simulations with digital Verilog netlist of digital blocks and block-level Spice netlists of analog blocks using AMS simulators.
 - Verification of the specification with simulation results.
- Full-chip-layout assembly
 - Collate all the layouts from all the block owners:
 - Analog block GDS from analog block owners.
 - Merged digital block GDS from digital block owners.
 - Full-chip assembly is done in Virtuoso: Import all the GDS files to DFII.
 - Populate the original floorplan with actual DFII data.
 - Finish the top-level chip connections.
 - Manual or automated.
 - Run full-chip verification (DRC/LVS/RX).
 - Stream out final GDS.

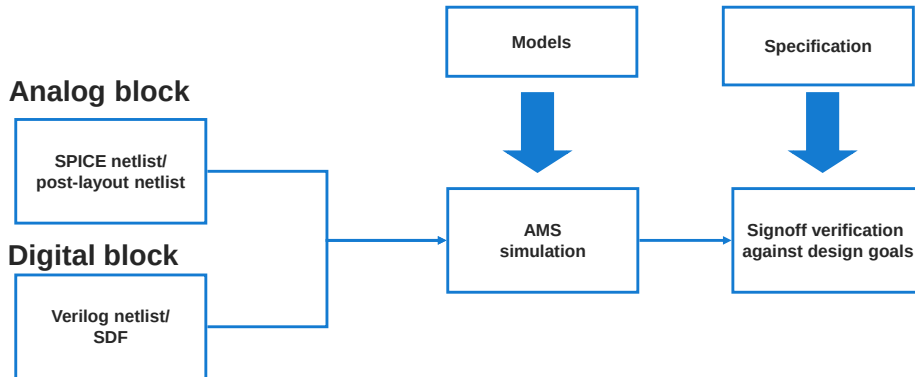


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Full-Chip Simulations (AOT)

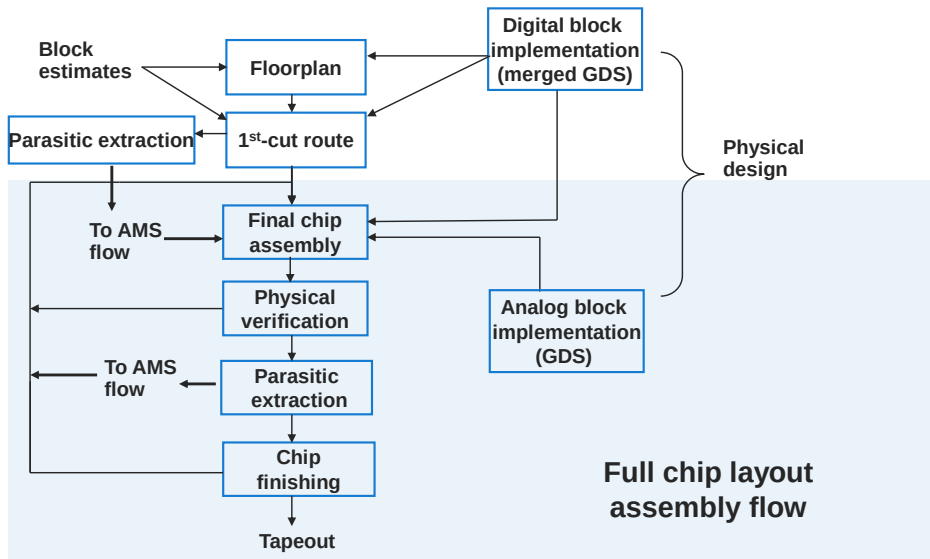
Full-chip-simulation groups perform all the digital and analog simulations to verify the circuit solutions, gate connectivity, and the sizes of the gates to meet system specifications.

This verification is performed using postlayout netlist at at the chip-finishing stage.



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Full-Chip-Layout Assembly (AOT)



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Full chip layout assembly:

- Collect all the layouts from all the block owners.
 - Analog block GDS from analog block owners.
 - Merged digital block GDS from digital block owners.
- Full chip assembly is done in Virtuoso: import all the GDS files to DFII.
- Populate the original floorplan with actual DFII data.
- Finish the top-level chip connections.
 - Manual or automated.
- Run full chip verification.
- Parasitic extraction and post-layout simulations.
- Chip finishes activities.
- Stream out final GDS.

Chip-Finishing Tasks: DOT

Chip-finishing tasks of DOT methodology comprise the following activities:

- Full-chip simulations
 - Full-chip AMS simulations with digital Verilog netlist of digital blocks and block-level Spice netlist of analog blocks using AMS simulators.
 - Verification of the specification with simulation results.
- Full-chip-layout assembly
 - Collate all the layouts from all the block owners:
 - Analog block LEF/GDS from analog block owners.
 - DEF/merged GDS of digital blocks from digital block owners.
 - Full-chip assembly is done in SoC Encounter: Import all LEF/DEF to Encounter.
 - Populate the original floorplan with the actual placed and routed database.
 - Finish the top-level-chip connections:
 - Automated.
 - Run full-chip verification at the merged GDS level of the full chip (DRC/LVS).
 - Stream out final GDS.



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Reading in the Full-Chip Floorplan

Reading in the floorplan file is the first activity of any full-chip owner during the chip assembly phase.

The following are some of the activities the chip designer has to be aware of during the chip assembly phase:

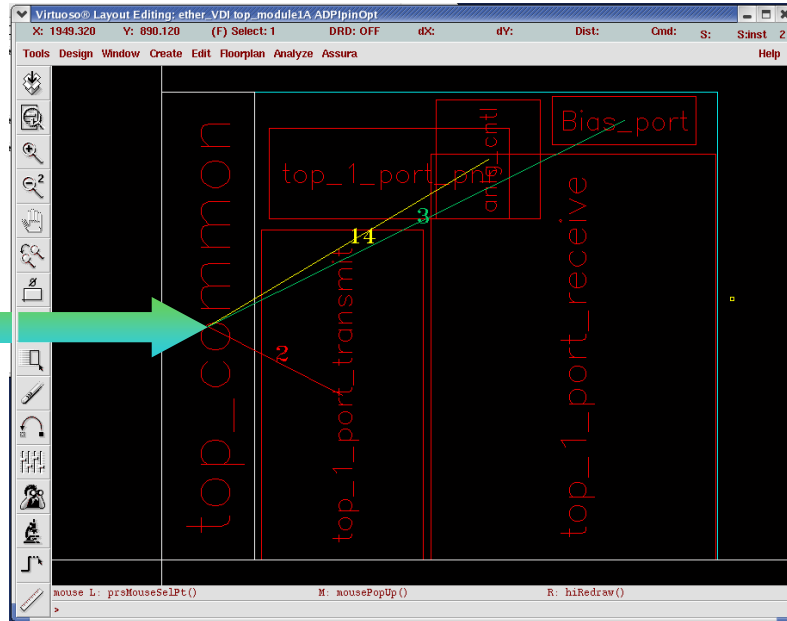
- Understanding different partitions or blocks of a design.
- Understanding the critical characteristics of each of the blocks – size, aspect ratio, and pins.
- Understanding the connectivity between blocks and connections to the pads.
- Finalizing the locations of the I/O pads to get the best connectivity among them.
- Allocating space for full-chip routing based on the number of routing layers and the number of unrouted pins and allocating space for needed substrate pick-ups.
- Allocate space for miscellaneous blocks and cells:
 - Seal ring, scribe line, and alignment cells.
 - Logo cells and revision cells.



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Full-Chip Floorplan View

The procedure to measure the total number of connections between the top_common block with others



Virtuoso preview or Virtuoso Chip editor is used for chip assembly activities.

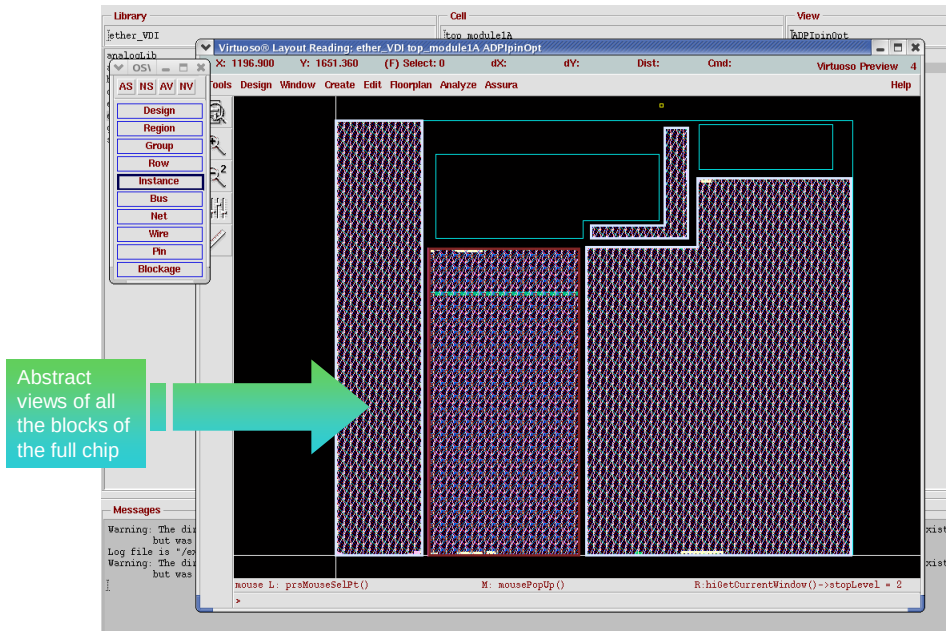
Update Floorplan with Actual Layout

- Whenever individual blocks in the chip are finished, you should replace the abstract view with the final block layout on the full-chip floorplan.
- The final floorplan might be affected by the accuracy of the block estimation.



Based on how accurate is the size estimation in the floorplan, this may affect the floorplan.

Full-Chip Floorplan View

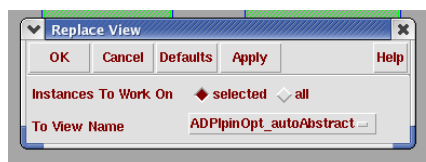
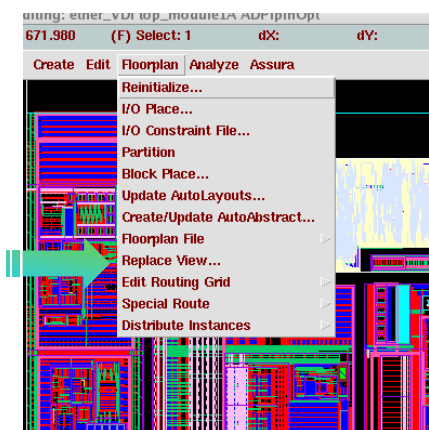


Working on abstract views during the full chip assembly/routing enables the tool to work faster rather than working on the actual layouts.

Actual layouts can be later populated as and when required during final physical verification.

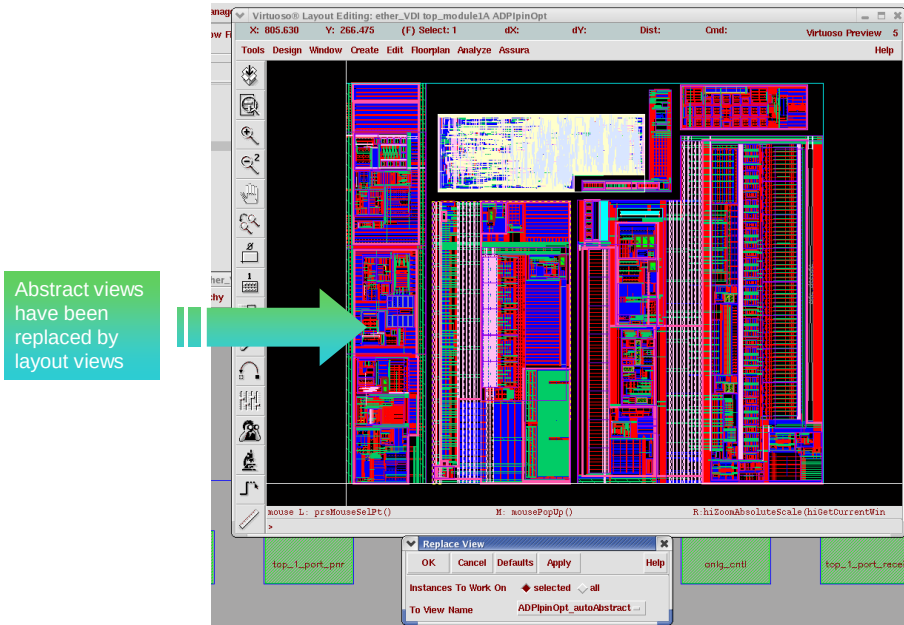
Replace Abstract Views with DFII Layout Blocks

Using an automated feature of Virtuoso Preview, either the selected or all block views from the abstract view can be replaced with the actual layout view of the respective block.



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Swap in Abstract Views with DFII Layout Blocks



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When all the layouts are replaced with actual layouts, layout editors run slower.

You can start using the World view option of the layout editor to navigate through the chip.

Full-Chip Routing

When all the blocks in the chip are finished, their abstract views are replaced with the final block layout on the full-chip floorplan.

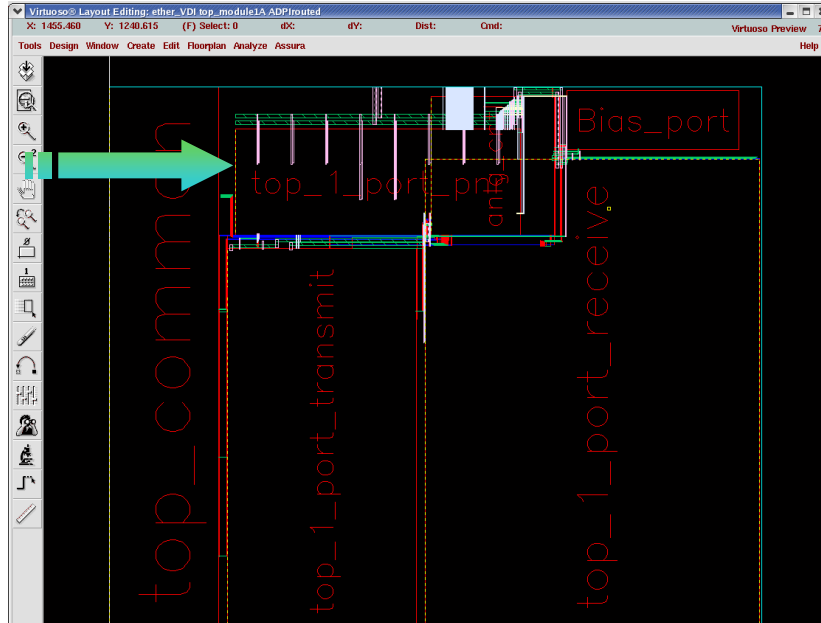
- In general, whenever there are 50 to 5,000 pins or nets, shape-based routers are the best tools for connecting them quickly with high quality.
- Routing of signals can be constrained by grouping them into classes, such as special groups or switching.
- User-defined routing rules for each layer and net-based-routing constraints address analog-signal crosstalk, minimum capacitance, and resistance effects.
- Routers
 - **Channel and maze router:** This router generally is used for digital place and route.
 - Pros: It simply connects pins and respects obstruction layers using simple paths on a coarse grid.
 - Cons: It does not have full features for routing constraints.
 - **Shape-based router:** This router generally is used for custom layouts.
 - Pros: It is very rich in features.
 - Cons: It has limited capacity. Usually, these tools are not a good choice if the design size has more than 200,000 nets.



A **channel router** is a specific variety of [routers for integrated circuits](#). Normally using two layers of interconnect, it must connect the specified pins on the top and bottom of the channel. Specified nets must also be brought out to the left and right of the channel but may be brought out in any order. The height of the channel is not specified - the router computes what height is needed.

Top-Level Analog Routing

Faster design time and high-quality top-level routing is achieved through automated routers like the Virtuoso Chip Assembly Router



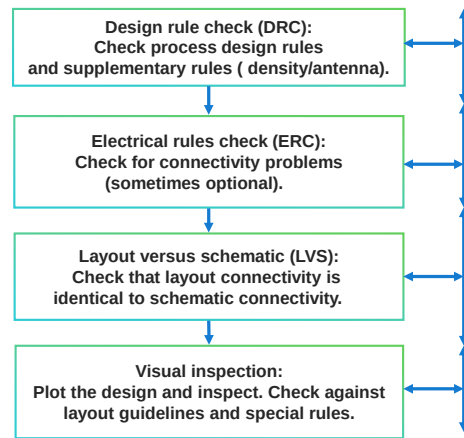
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VCAR → Virtuoso chip assembly router.

Physical Verification Flow

- **Layout Verification** is a process used to check the layout on design rules and the connectivity of the layout which directly ensures the correctness of the circuit.
- With the growing design sizes, designers and tool providers have developed methodologies to verify higher numbers of transistors.



Why do we need plotting at all?

One reason is that there may not be coding for all of the rules for DRC verification. The second is that in some cases, some rules are very rare, and the DRC rule check may not be deterministic. The third important reason is that architectural improvements may only be understood from a visual inspection of a large-scale plot.

Preparing for Chip-Level Verification

To handle large chips with a huge transistor count, designers developed a robust verification methodology that involves a phased approach to achieve chip-level clean verification.

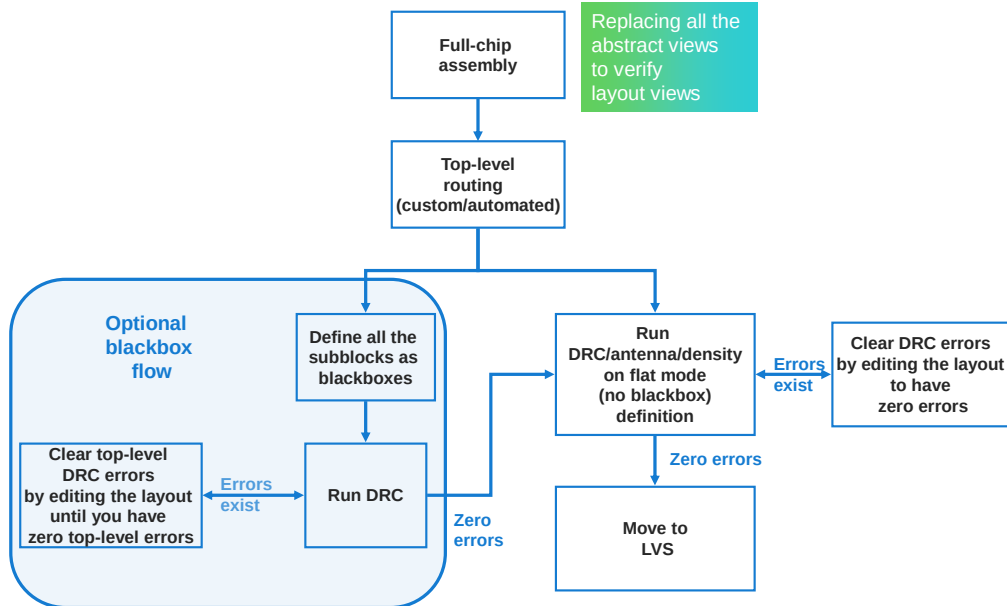
The following are some steps of this phased approach.

- All the blocks should be LVS/DRC clean at the block level before being verified at the chip level.
- After the top-level assembly and routing are done, check for these:
 - Top-level connectivity by running LVS in blackbox mode.
 - Top-level DRC/antenna errors by running DRC in blackbox mode.
- Ignore the following cells during full-chip DRC/LVS runs:
 - Alignment cells
 - Logo cells
 - Revision cells
- If the chip is clean in blackbox mode, then start working on flat full-chip-level verification.



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Full Chip Verification Methodology: DRC

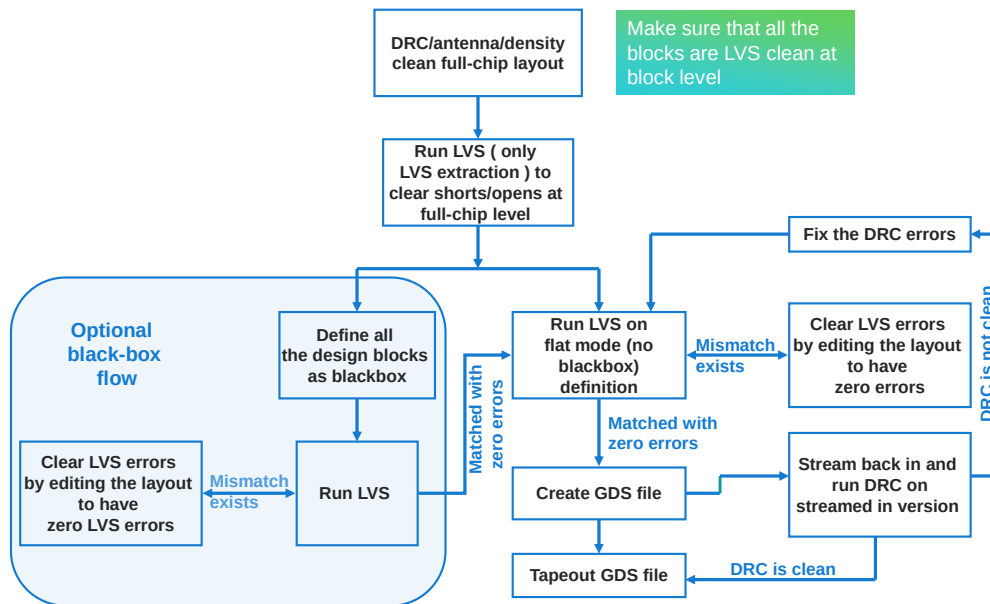


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- Full chip assembly is a process where you have all your blocks implemented and have them verified to all the block-level constraints.
- Blackbox flow: This flow is followed to make sure that the top-level routing + placement correctness once the blackbox flow is clean.
- Full chip flow is a complete flow that makes sure that all the connectivity and manufacturability.

Full Chip Verification Methodology: LVS



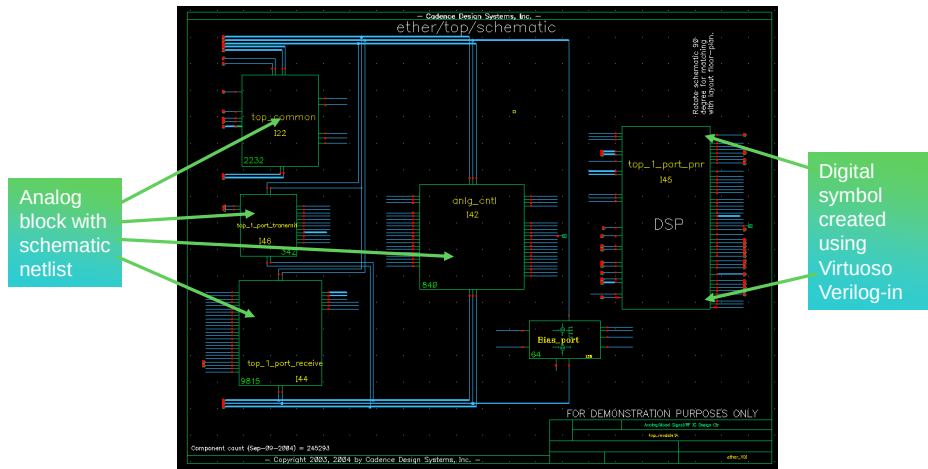
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- Full chip assembly is a process where you have all your blocks implemented and have them verified to all the block-level constraints.
- Blackbox flow: This flow is followed to make sure that the top-level routing + placement correctness once the blackbox flow is clean.
- Full chip flow is a complete flow that makes sure that all the connectivity and manufacturability.

Full Chip Schematic Reference

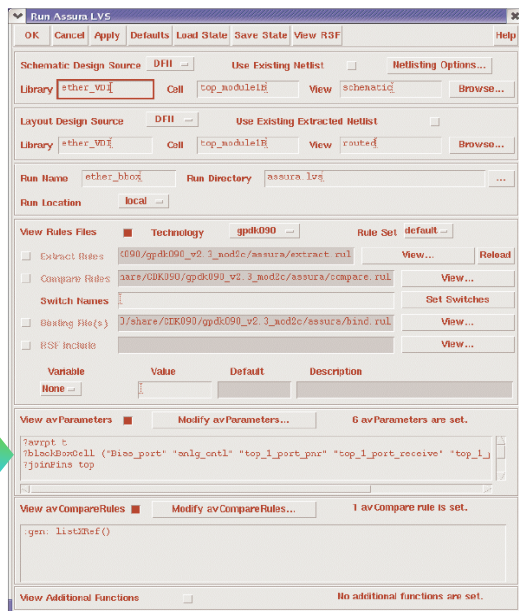
- This schematic is used as a reference netlist during LVS verification of the full chip.
- Analog blocks are derived from drawn schematics and digital blocks are derived from Verilog-in from a place and routed Verilog netlist.



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Blackbox Full Chip Verification: LVS

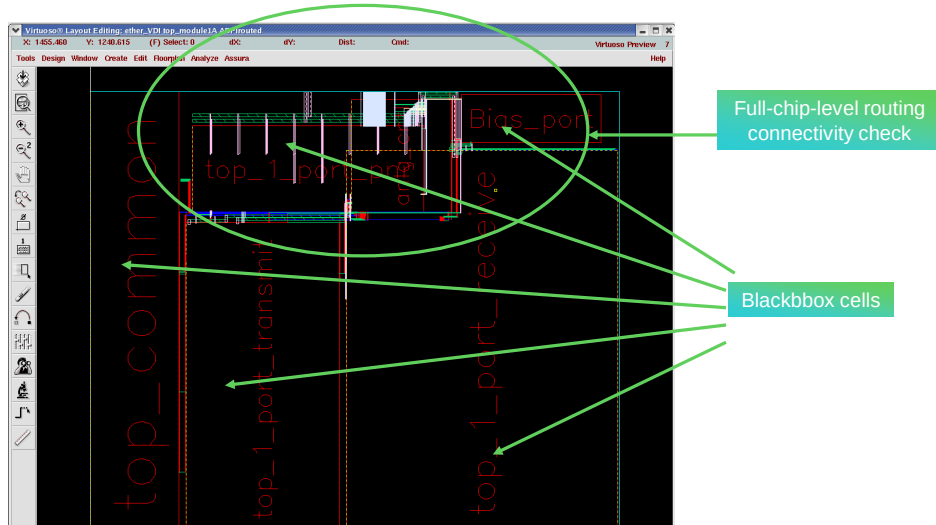
Blackbox
definition
during
Assura LVS



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Blackbox LVS Verification: Full Chip

Top level connectivity check is performed by doing a black-box verification of all the design blocks.



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Subsection

Final Physical Verification (DRC, LVS, Antenna)

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Final Physical Verification

- Final physical verification is the last activity before taping out the design.
- During this step, verification is performed at the flat level to make sure that flat GDS is the same as the reference circuit netlist.
- This approach does not include any blackbox flow.
- As the verification is done at the flat level, this approach involves large computing resources to reduce the verification run times for which the present physical verification tools support the following technologies:
 - Multithreading, multiple-processor support
 - Multiple CPU machines
 - Server farm support (LSF farms)
- Finally, to double-check the full-chip layout, a visual inspection is done using the layout plot of the chip with the help of several experienced designers.

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Full chip owners are generally very cautious against last-minute surprises, due to which they perform complete flat DRC/LVS checks to make sure the GDS is correct before sending it to the FAB.

Mask designers start their job from design GDS:

- Once the full chip GDS is clean with physical verification.
- The final GDS is fractured into several files and sent to the FAB.

Layer fracturing:

- Layer fracturing is a process where you break the complete chip layout into separate layers and send it to the Mask Shop.

Mask shop takes this GDS as input and starts creating masks.

Final Physical Verification (continued)

The
flattenCell-
Contents
function
enables a
flat full-chip
DRC run

Run Assura DRC

OK Cancel Apply Defaults Load State Save State View RSF Help

Layout Design Source **DFII** Compare two layouts ☐ Generate Lvl Compare Rules...

Library **ether_VDI** Cell **top_module1A** View **ADP1layout** Browse...

Save Extracted View ☒ View Name **drc_extracted**

Area To Be Checked **Full**

Run Name **top_pnd** Run Directory **assura.drc** ...

Run Location **local**

View Rules Files ☒ Technology **gpdK090** Rule Set **default**

☐ Rules File **DK090/share/CDK090/gpdK090/assura/drc.rul** View... Reload

Switch Names Set Switches

☐ RSF Include View...

Variable	Value	Default	Description
None	1		

View avParameters ☒ Modify avParameters... 2 avParameters are set.

?avprt t
?flattenCellContents ("top_module1A")

View Additional Functions ☒ 1 additional function is set.

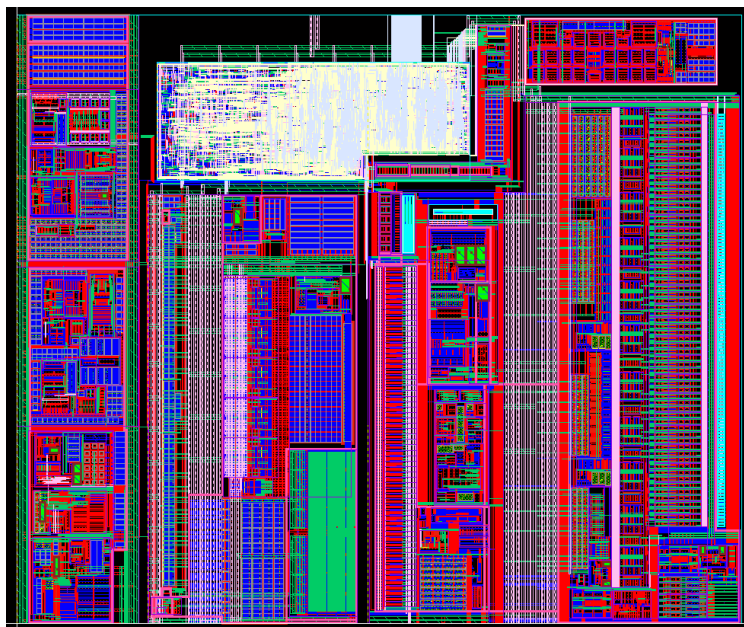
☐ Use avFlattenCell Function ☐ Use joinableNet Function

☒ Modify New Database Options... ☐ Use changeLabel Function



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Final Physical Verification (continued)



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Subsection

Chip-Finishing Tasks

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Chip-Finishing Tasks

Chip-finishing tasks include the verification of all devices, implementing a special class of cells (mentioned below), and ensuring compatibility with the manufacturing process.

- Examples of chip-finishing cells include the following:
 - Revisit focused ion beam (FIB) cells.
 - Put identification marks on chips, such as logos, design revision and copyright markings. These marks should be big enough to see with the naked eye.
 - Put probe pads on necessary signals for debugging purposes.
 - Put alignment keys for processing equipment. This is not required with most of the fabs. Usually, you should consider this when you are dealing with internal fabs.
- Stream-out then stream-in and run LVL on the streamed-in version versus the original layout.



Though these tasks are performed by the data preparation team.

At a minimum, the layout designer should receive all of this information ahead of time, to allocate space for the required devices; otherwise, there is a danger that there will not be enough free space on the chip to implement them.

The chip-finishing task does not end when all the devices that are responsible for the functionality of the chip are placed and verified; there is still work to be done in implementing a separate class of cells to finish the chip and ensure that the chip is compatible with the manufacturing process.

Chip-Finishing Tasks (continued)

- The layout designer should receive all this information ahead of time, to allocate space for the required devices and to avoid any surprises at the end.
- It is important to note that after all the devices are placed and verified individually, the final full-chip verification should include all of these chip-finishing cells which in turn will be included in mask making.
- Be aware that placing an identification mark in a wrong place might cause an electrical problem.



- The layout designer should receive all this information ahead of time, to allocate space for the required devices, to there may be a potential da that there will not be enough free space on the chip to implement them.
- It is easy to cause an electrical problem by placing an identification mark in the wrong place.

FIB Cells

FIB cells are cells added during the chip-finishing flow or earlier in the subblock layout, near the critical design blocks of the chip to facilitate circuit-adjustment options for the post-manufactured ICs.

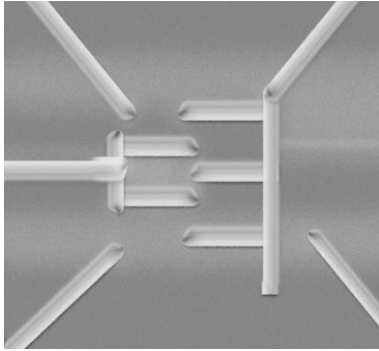
- Typically, FIB cells are inserted for the following reasons:
 - A spare buffer FIB cell is placed near a critical clock routing so that any critical clock-timing failure can be adjusted or analyzed with this cell.
 - FIB cells are used in the analog portion for design adjustments during prototyping of a design.
- FIB cells are generally implemented in polysilicon or metal and must be built in such a way that a FIB machine can accurately blow them out.
 - The FIB cells must be specially designed to isolate the impact on the rest of the internal circuitry because of laser zapping or tungsten deposition.
- The number of FIB options should be optimized so that the repair time can be minimized.



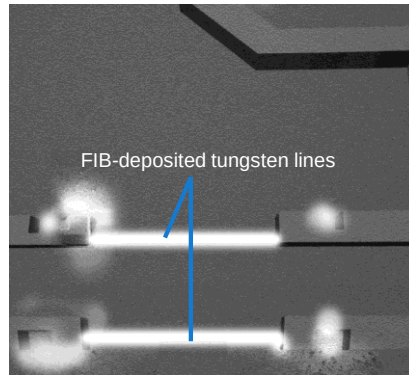
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FIB Cells (continued)

- Using this technique, a circuit can be modified to rewire or repattern certain interconnections.
- The effect of each modification on the functionality of the chip can be observed.



Typical FIB cell which facilitates both rewiring and cutoff.



Through backside milling of the substrate, the desired interconnection is made with tungsten.

FIB micromachining is to rewire or re-pattern certain interconnections or cut off certain interconnections on real chips to realize IC chip function changes. These techniques are very helpful for IC designers to debug their design errors on testing chips before mass productions.

Alignment Keys

Alignment keys are special structures that are inserted during the chip-finishing flow to enable precision optical alignment during fabrication.

There are many alignment keys in an integrated circuit. They vary depending on the process and manufacturing requirements. Here are a few.

- An alignment key is made of various layers that can be seen from the chip level. Metal and via layers are typical choices. 
- NIKON keys are used to align the reticle when generating the masks on the wafer.
 - Every layer is placed in this key.
 - These keys are considered references for the optical lithography process. 
- As these cells are part of the full-chip GDS, they never pass layout verification because they generate design rule errors and illegal devices.
 - NIKON keys need to be placed in all four corners of the chip and as close to the corners as possible.



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Probe Pads

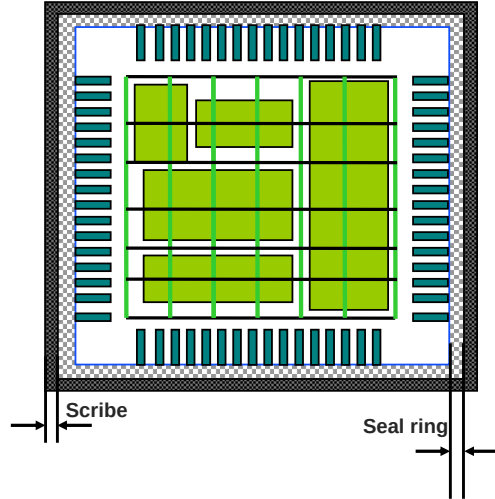
- In the first revision of the die, the designer usually will need to debug potential problems. So, the designer must have access to the specific nodes of the layout.
- For access, there should be a *probe pad* on top metal. The purpose of the probe pad is to provide a site for a probe tip to read the voltage or monitor the signal of that specific node.
- Circuit designers should use an optimal number of probe pads for debugging the circuit.
- For example, having $10\mu \times 10\mu$ top metal square pad enables the designer to use a probe station, put a probe tip on the pad, and read the voltage value.
- Using probe pads is common in the fabrication stage also. For example, with larger probe pads, fuse trimming can happen at the fabrication site. For example, fuse trimming can be used for bandgap or offset trimming.



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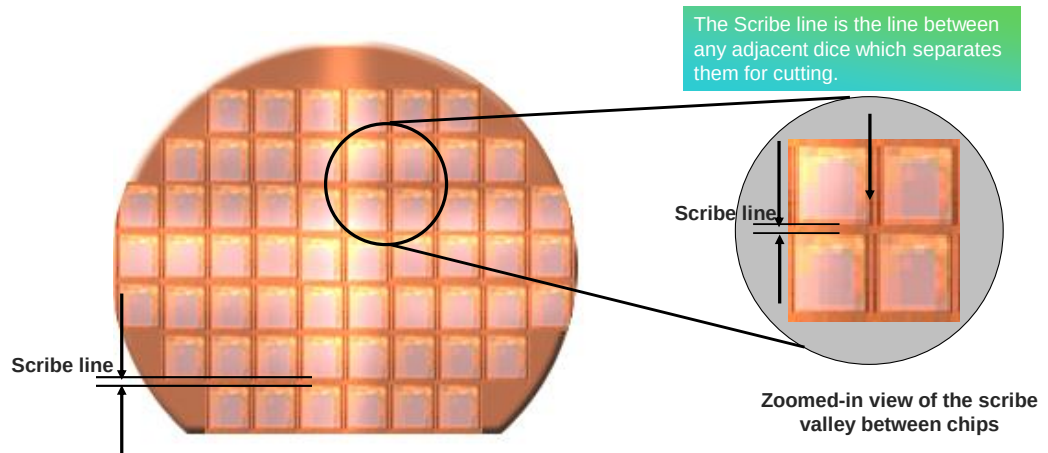
Scribe Line

- A large slice of silicon is called a wafer. After the wafer is manufactured, the wafer is diced into individual ICs.
- Narrow channels between individual ICs are mechanically weakened by scratching them with a diamond tip. These channels are known as the *scribe lines*.
- The wafer is cut along the scribe with a diamond blade or burnt with a laser. The wafer is then mechanically stressed and broken apart along the channels, thereby separating the individual ICs.



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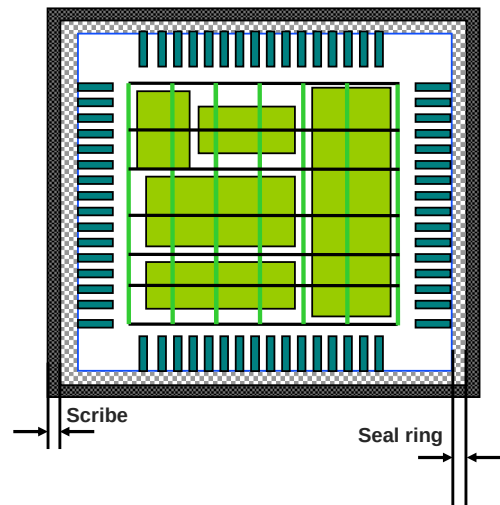
Scribe Lines of a Chip



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Seal Ring

- In order to protect the IC during the die-cutting process, a *seal ring* is required.
 - The seal ring is implemented around the edge of the chip.
 - It has all the layers of process such as diffusion, poly, metal, and so on.
- The seal ring protects the logic circuitry against the contaminants that are produced during the die-cutting procedure.
 - When the scribe lines are scratched to weaken the joints, the contaminants can cause physical damage to the manufactured die.
 - The seal ring protects the logical circuitry from both mechanical and electrical disturbances.



It is easy to cause an electrical problem by placing an identification mark in the wrong place.

Logo and Identification Mark

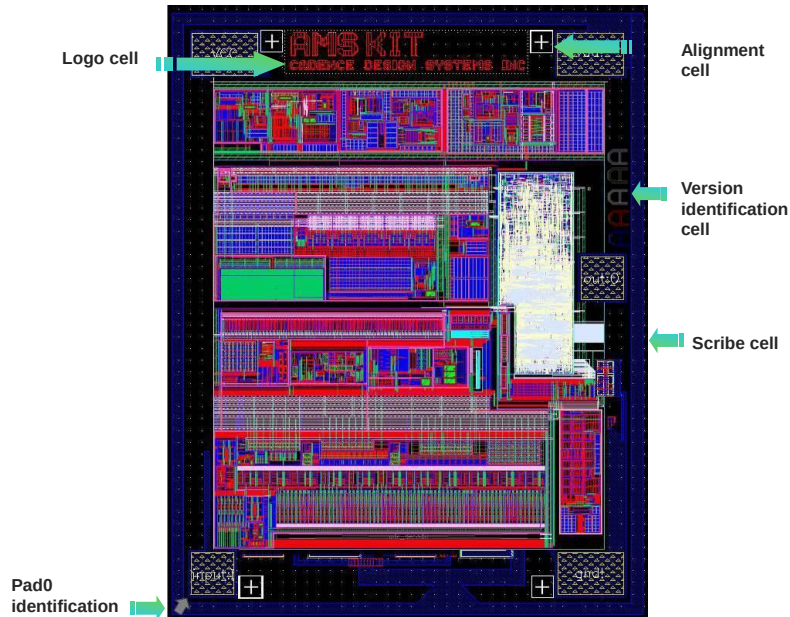
A **logo** and **identification mark** should be added to the layout to show the ownership, name, and number of the die. The mark should be visible to the naked eye.

- Other than this identification mark which usually shows the revision of the die, the layout engineer adds a letter notification for all the layers.
 - For example, if the process has 24 layers (such as different layers of metal, poly, N+, P+, and so on) then there will be 24 A letters somewhere on the die in the first revision of the die.
 - In the next revision if only metal layers are being changed, then the letter for the metal layers will be B to show that it is a second revision and the letters for the other layer will remain A.
- The revision letters do not pass DRC, and these DRC errors should be waived.



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Full Chip Assembly



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All these miscellaneous cells have to be ignored during the full chip DRC/LVS verifications.

Other Chip-Finishing Tasks

- Full-chip owners are responsible for several activities:
 - Full-chip assembly
 - Full-chip routing
 - Insertion of miscellaneous cells
 - Full-chip verification
- After all these activities are done by many designers, there is a definite possibility of manual errors. To correct these errors, a visual review or technical audit must be done.
 - An effective audit uses a specific and detailed checklist and has an auditor.
 - An auditor is someone who understands the extent of the problem and can propose solutions.
 - In general, the auditor should be a person who has not participated directly in the project so that any bias to design styles and methodologies is avoided.



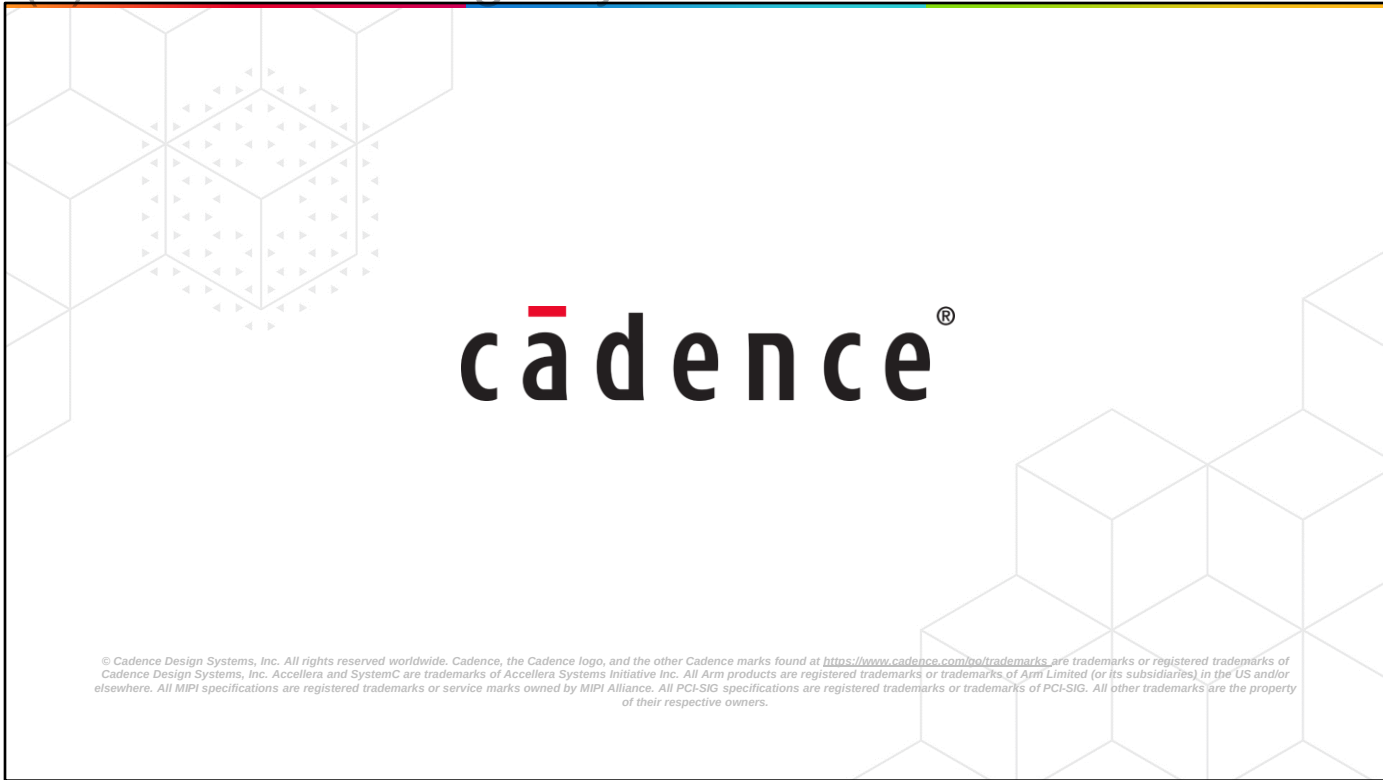
The secret to performing an effective audit is to use a specific and detailed checklist and to have as an auditor a person who understands the extent of the problem and can propose solutions. In general, the auditor should be a person who has not participated directly in the project so that any bias to design styles and methodologies is avoided.

Full Chip Layout Checklist: Example

- Is the power-supply strapping adequate and implemented with enough vias? ☐ Yes ☐ No ☐ N/A
- Is the length of all critical path signals optimized? ☐ Yes ☐ No ☐ N/A
- Have all special signal requirements been satisfied? ☐ Yes ☐ No ☐ N/A
- Is the full-chip DRC/LVS/antenna clean? ☐ Yes ☐ No ☐ N/A
- Has the DRC waiver file been verified from the verification tools ?
☐ Yes ☐ No ☐ N/A
- Are there probe pads for specified signals? ☐ Yes ☐ No ☐ N/A
- Is there any sensitive circuitry placed close to the edge of the die?
☐ Yes ☐ No ☐ N/A
- Are the rules for the chip corner areas satisfied? ☐ Yes ☐ No ☐ N/A
- Is the interface of the chip to the seal ring properly defined?
☐ Yes ☐ No ☐ N/A
- Have all ESD and latchup requirements been satisfied? ☐ Yes ☐ No ☐ N/A
- Have all the necessary chip-finishing cells been included? ☐ Yes ☐ No ☐ N/A
- Is the origin in the center of the die? ☐ Yes ☐ No ☐ N/A



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